

Nilarnab Debnath

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EDUCATION

Examination	Institute	Graduation Year	CGPA / %
Bachelor of Technology	Malaviya National Institute of Technology Jaipur	2023	9.24
12 standard	Sacred Heart Children Senior Secondary School, Kota, RJ	2019	91.8 %

RESEARCH EXPERIENCE

A Visual Question and Answering System with Support for Compound Emotions Using Facial Landmark Identification with MediaPipe and CNN Classifier • Published in *Neurocomputing, Elsevier* ([link](#)) July'24

- Enhanced the scope of visual question and answering by adding support for emotion identification.
- Worked on compound emotion identification by facial landmark identification using MediaPipe and CNN classification.
- Used attention-based encoder-decoder model to generate caption to get the basic context of the image.
- Employed RetinaFace for face identification and Deepface for simple emotion identification.
- Applied BERT to answer questions based on the generated context of the image.
- Implemented a modular visual question-answering system, allowing independent and parallelizable training.
- Increased compound emotion identification accuracy to 44%, improving upon traditional classifiers, like VGG19's 33%.
- Boosted emotional question and answering accuracy to 74%, improving upon the state-of-the-art One-For-All model's 64%.

Bilingual Question and Answering System Given A Set of Known Questions using Deep Siamese Neural Networks, XGBoost Machine and DistilBERT (ongoing)

- Developed a question-answering system prioritizing responses from known questions before answers from DistilBERT.
- Implemented paraphrase identification to get a similar known question of an unknown question.
- Used Siamese Neural Network for English and XGBoost for Bengali paraphrase identification.
- Achieved 75% accuracy in English and 95% accuracy in Bengali paraphrase identification.

PROFESSIONAL EXPERIENCE

Member of Technical Staff, Salesforce

July'23-Now

Application Performance Monitoring Using Opentelemetry

- Worked on integrating Opentelemetry with salesforce monitoring cloud architecture for increasing visibility of services.
- Added event processor by overriding methods of SpanProcessor class to transform them into salesforce-specific event format.
- Modified SpanProcessor to send spans in a salesforce-specific format so that they can be ingested by the downstream services.
- Exposed API for custom metrics by mimicking and modifying Opentelemetry SpanBatchProcessor.
- Implemented enrichment of child span with root span name by creating trace cache.
- Received accolades for adding customization options to the users and on-time delivery of features.

Cloudwatch Log Exporter Service

- Streamed, processed, and exported CloudWatch logs with AWS Kinesis, to Salesforce's monitoring backend via a hosted log exporter.
- Worked on Terraform and Spinnaker to automate the provisioning of AWS resources and deployment of the service in the cluster.
- Received appreciation for enhancing the onboarding experience and increasing the scope of existing customers to teams with no clusters.

Government Cloud Compatibility Enhancement of Cloudwatch Log Exporter

- Identified and Resolved AWS ARN naming issue that blocked service functionality in government clouds due to differing conventions.
- Modified the helm chart to allow flexibility for government cloud-specific deployments through spinnaker deployments.
- Earned recognition for quick fixes that required no changes from the customers' side.

Software Developer Summer Intern, Salesforce

June'22 - Aug'22

Dead Letter Queue Implementation for Telemetry Handler

- Worked on handling dropped telemetries by adding dead letter support on a spring boot application.
- Used AWS SQS services to come up with a pipeline that facilitates retry in case of valid telemetries.
- Worked on thread pools in spring boot to achieve concurrent execution.
- Implemented exponential back-off for retrying valid telemetries to handle drops during downstream service downtime.
- Increased visibility of dropped telemetries for both the team and the customers' side.
- Received appreciation and a Pre-Placement-Offer for successful design and timely implementation.

Software Developer Summer Intern, PillarPlus Pvt Ltd

May'20 - Jul'20

Architectural Design Automation Using Python Scripts

- Worked on plan and elevation design automation using Python libraries, and concepts of coordinate geometry.
- Used ezdxf and NumPy to automate drawing of electrical and plumbing lines in plan and elevation.
- Expedited architectural drawing for engineers by automating block detection and identification.

ACADEMIC AND PROFESSIONAL ACHIEVEMENTS

Secured First position in Customer-Centric Hackathon by Salesforce

- Designed and implemented a grievance-aware AI-based adaptive feedback system
- Categorized grievances using Einstein GPT embeddings and used Agglomerative Clustering algorithm.
- Allowed a plug-and-play model with Slack API to set sources of grievances to any channel easily.
- Implemented adaptive prioritization of questions to get the most insight from partial responses from customers.
- Used Einstein-LLM to dynamically generate questions from the grievance categories.
- Leveraged Lightning Components to make the solution compatible with salesforce architecture.

Secured Second Position in the Web Design Competition Organized by College Information Communication Technology Center

- Designed a basic, multiple-screen compatible, animated, and robust system using HTML, CSS, JavaScript, and Bootstrap.
- Applied neumorphism using box-shadow properties to create an embedded GUI effect;
- Implemented glassmorphism to produce a frosted-glass effect.
- Used CSS and JavaScript to create interactive and visually engaging animations.
- Leveraged php-ml to perform supervised classification using KNN.

ACADEMIC PROJECTS

Video Streaming Based E-Commerce React Native Application

- Used React Native to develop a robust, cross-platform mobile application with express js API as its backend.
- Implemented Dynamic Adaptive Streaming Over HTTP to automatically change video quality on internet speed fluctuations.
- Used Minio as an affordable substitute for AWS S3 for storage of video and image files.
- Deployed Meilisearch to enable product searches by name, description, or promoter influencer.
- Integrated Firebase Mobile Authentication for otp-based login and geolocation for delivery address identification.
- Configured Apache as an API Gateway to route requests of different app versions to their compatible API instances.
- Integrated Razorpay API to allow seamless payment functionality.
- Achieved year-long funding from the college innovation and incubation center for a better shopping experience and consistency in work.

Security Vulnerability Demonstration Using Key-logger and Real-time Data Integration

- Identified a security vulnerability in on-campus computers using a Python-based key-logger.
- Demonstrated potential system infiltration without admin access using Windows 10's inability to detect trojan horses in a zip file.
- Integrated Firebase real-time database with the key logger for easy collection and retrieval of the data.
- Earned recognition from professors for uncovering critical security flaws, and driving improvements in system safeguards.

Hostel Management Website with Laravel PHP ([link](#))

- Used PHP and MySql for the backend, database management, HTML, CSS, JavaScript, and Bootstrap for the frontend.
- Implemented Laravel Auth module to make authentication module development fully automated.
- Used Ajax to create asynchronous calls to save reloading time.
- Made hostel room, mess menu, and food supply management easy and automated.

HR Work Automator and Event Assistor Website

- Used PHP and MySql to maintain and update attendance databases in the portal.
- Implemented an animated, robust, interactive, and user-friendly interface using HTML, CSS, javascript, bootstrap
- Developed multiple access levels and added mailing features to alert users regarding each event.
- Used Ajax to efficiently update the page and FPDF to generate certificates.
- Made event organization easier and created a fundraiser's database.

Python Command Line Application for Finding Open Slots in Vaccination Drive

- Used Python requests library to fetch raw data of vaccination centers from *setu API* and convert them into JSON.
- Traversed data to find if there is an available slot for a given age limit and geographical location.
- Ran the program every 5 seconds by cron and a beep was produced by Winsound if a slot would open at any given time.
- Made slot booking easier as multiple manual checking was rendered completely redundant with this application.

TECHNICAL KNOWLEDGE

Languages: Python, C, C++, Java, PHP, MySql, HTML5, CSS3.

Concepts: APM, Kubernetes, Web development, RDBMS, Web scraping, k-NN.

Tools: K9s, Splunk, Terraform, Spinnaker, APM agent, Firebase, Git, Android Studio, PhpMyAdmin.

Libraries: Tensorflow, OpenCV, Opentelemetry, ezdxf, Python Imaging Library, Pyautogui, pandas, NumPy, Webbrowser, keyboard.

CO-CURRICULAR

- Hope of People Everywhere (NGO): As a lead of the Fundraising vertical, worked in collecting funds for underprivileged children.
- College Coding Club Mentor: Worked as a mentor and organized coding contests and online web development workshops.
- Painting Competition: Received first prize in an on-campus canvas painting competition.